

# OpenFOAM Mesh Refinement Analysis

Brayden Falla

# Why Pivot?

- Contribution was planned and outlined
- Original post made in February, resolved late April
- Not every community will hand hold

 Activities

 **henry**  
© 2026-04-17 18:02  
manager 🔍 -0013835

Resolved by commit a21579d3b75cbb02599f34882d4f65b7a6f1e261

| 🔄 Issue History  |          |                     |                 |
|------------------|----------|---------------------|-----------------|
| Date Modified    | Username | Field               | Change          |
| 2026-04-12 07:02 | bafalla  | New Issue           |                 |
| 2026-04-17 18:02 | henry    | Assigned To         | => henry        |
| 2026-04-17 18:02 | henry    | Status              | new => resolved |
| 2026-04-17 18:02 | henry    | Resolution          | open => fixed   |
| 2026-04-17 18:02 | henry    | Fixed in Version    | => dev          |
| 2026-04-17 18:02 | henry    | Note Added: 0013835 |                 |

View Issue Details

Jump to Notes

Jump to History

| ID               | Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Category    | View Status | Date Submitted   | Last Update      |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|------------------|------------------|
| 0004327          | OpenFOAM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Bug         | public      | 2026-02-23 17:10 | 2026-04-17 18:02 |
| Reporter         | Tobermory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Assigned To | henry       |                  |                  |
| Priority         | normal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Severity    | text        | Reproducibility  | always           |
| Status           | resolved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Resolution  | fixed       |                  |                  |
| Platform         | GNU/Linux                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | OS          | Other       | OS Version       | (please specify) |
| Product Version  | 13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |             |                  |                  |
| Fixed in Version | dev                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |             |                  |                  |
| Summary          | 0004327: User Guide still refers to printCoeffs ... but I think that this has been removed?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |             |                  |                  |
| Description      | <p>The online User Guide (<a href="https://doc.cfd.direct/openfoam/user-guide-v13/turbulence">https://doc.cfd.direct/openfoam/user-guide-v13/turbulence</a>) still talks about the optional printCoeffs flag in the momentumTransport file, to allow confirmation of the model coeffs. I think* that this needs removing, since all of the coding around printCoeffs has been removed in v13 (cf v8), as far as I can see (ie from RASModel.C and the individual RAS models). Happy to be proven wrong.</p> <p>Certainly, setting the printCoeffs flag does nothing. Which begs the follow up question - how DO you print the coeffs now, or do you just need to go back to the source?</p> |             |             |                  |                  |
| Tags             | No tags attached.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |             |                  |                  |

Monitor

# Intro to PreCICE

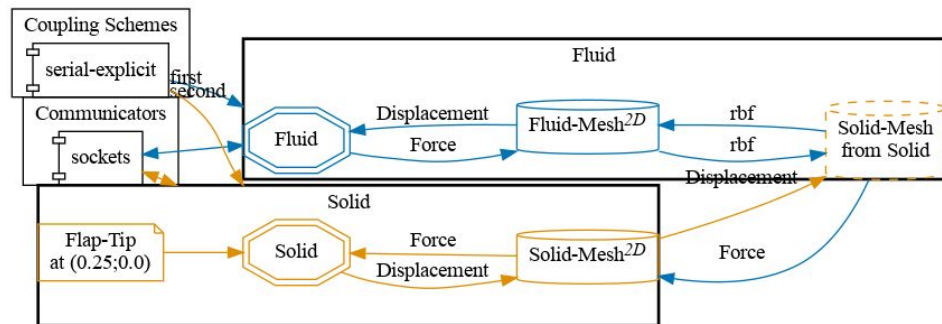


- Open-source fluid-structure coupling tool
- Utilizes individual adapters that are specially compiled and run in tandem
- Has suite of examples set up for coupling

```
source $HOME/software/OpenFOAM/OpenFOAM-dev/etc/bashrc
source /usr/lib/openfoam/openfoam2512/etc/bashrc

export SU2_RUN=/home/april/software/SU2_RUN
export SU2_HOME=/home/april/software/SU2-7.5.1
export PATH=$SU2_RUN/bin:../su2-adapter/run:$PATH
export PYTHONPATH=$SU2_RUN/bin:$PYTHONPATH

bind 'set bell-style none'
export PATH=/home/april/software/su2-adapter/run:$PATH
```



# Solid Mechanics - FEniCS

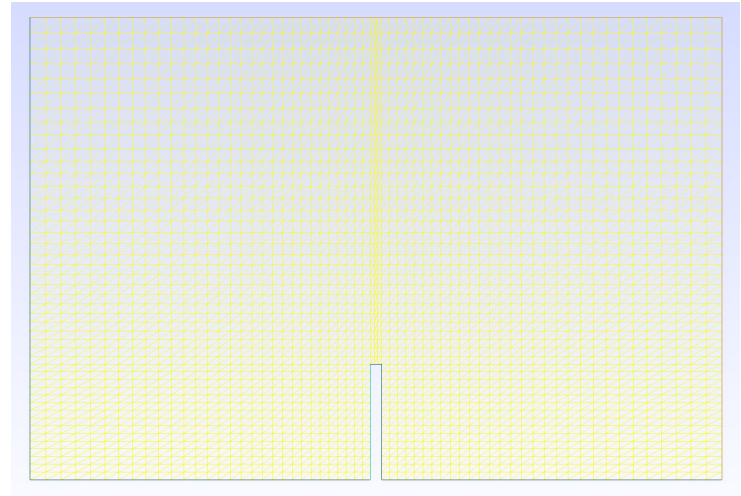
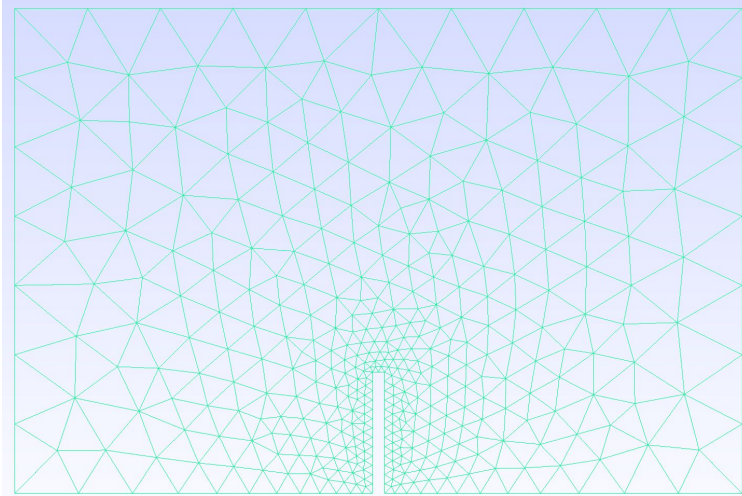
- Open-source platform/PDE solver for FEM.
- Python wraps around compiled C++ libraries
- Mesh generation/problem setup specified largely in Python script
- Hosted on GitHub
- Started in 2003 at University of Chicago



# Fluid Mechanics

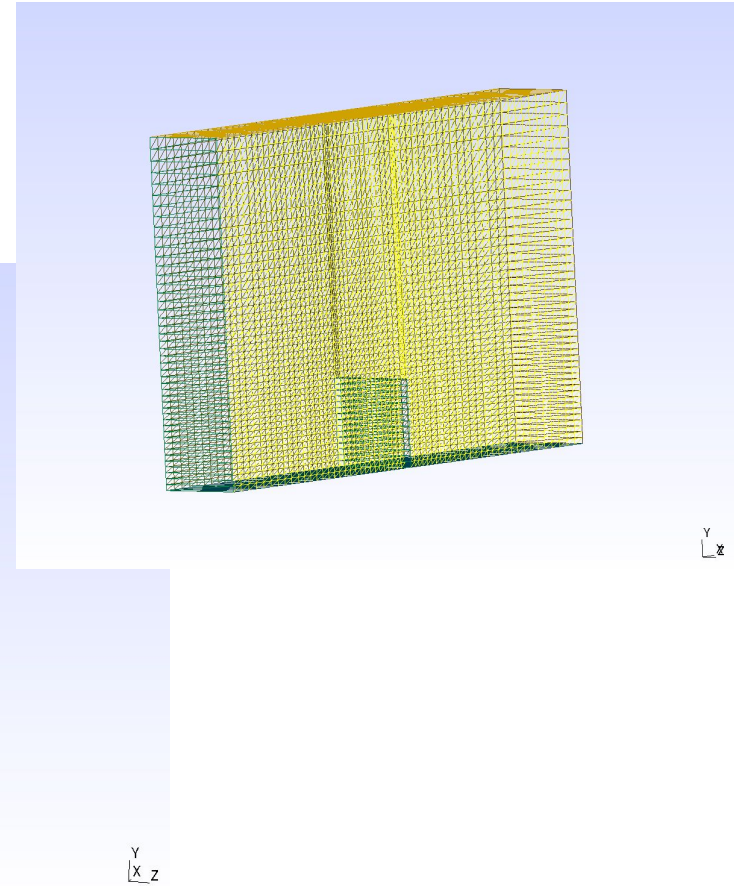
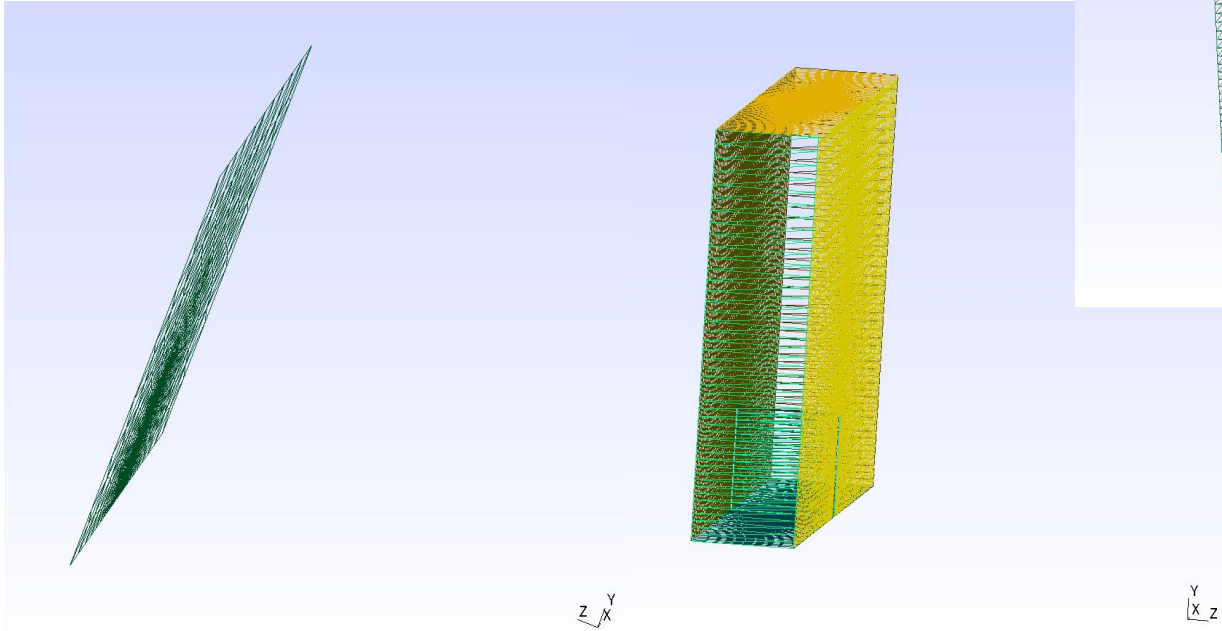
# Open $\nabla$ FOAM

- SU2 without source mesh file was non-viable
- OpenFOAM chosen as original focus as well as mesh flexibility
- PCG vs Euler Solvers

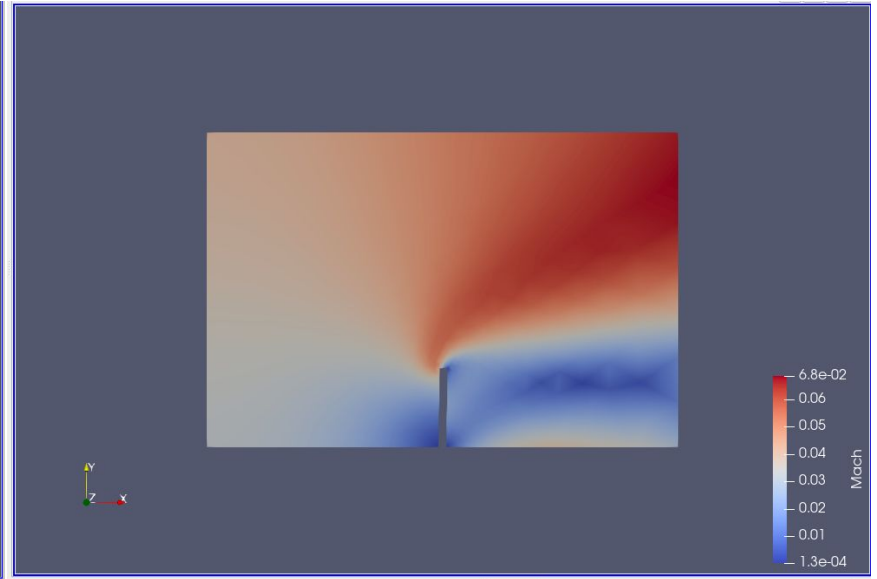
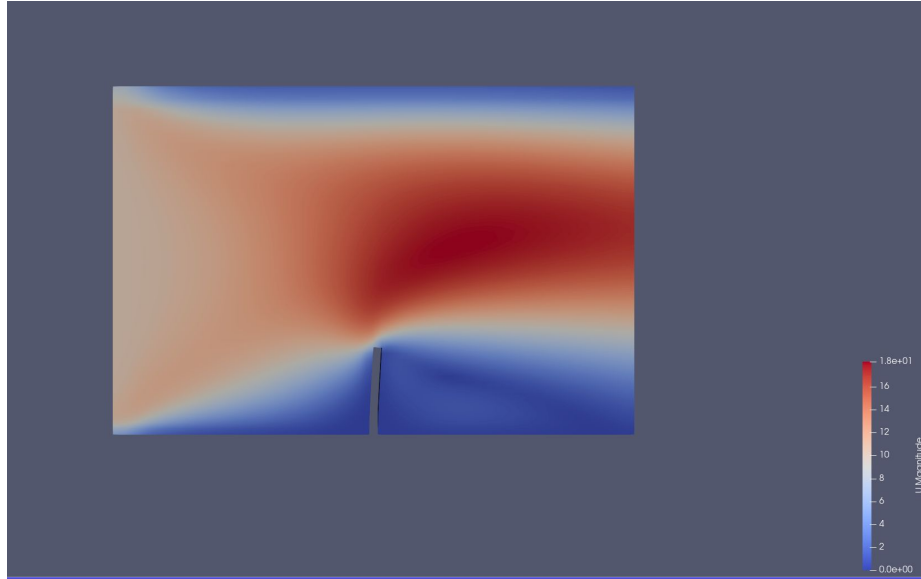


# More Differences

- True 2D vs projected 3D

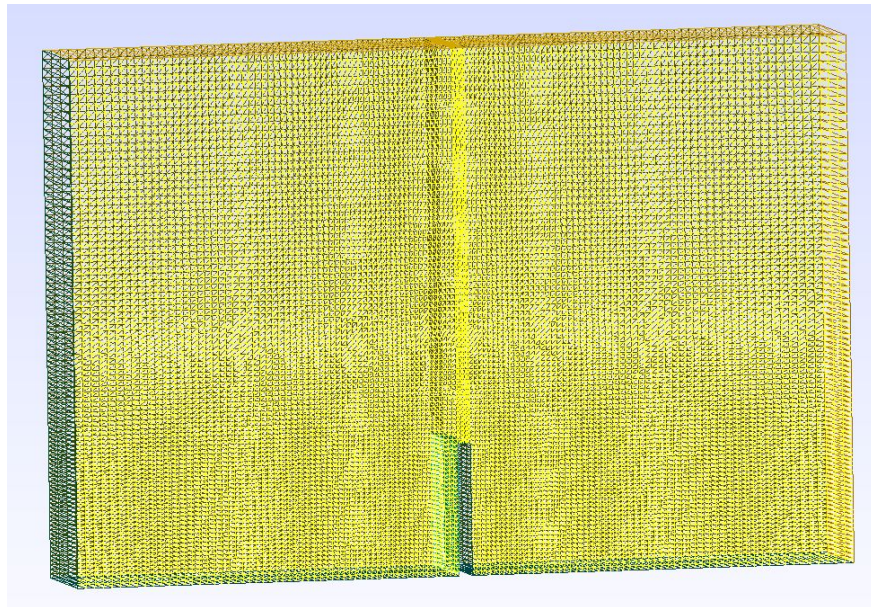
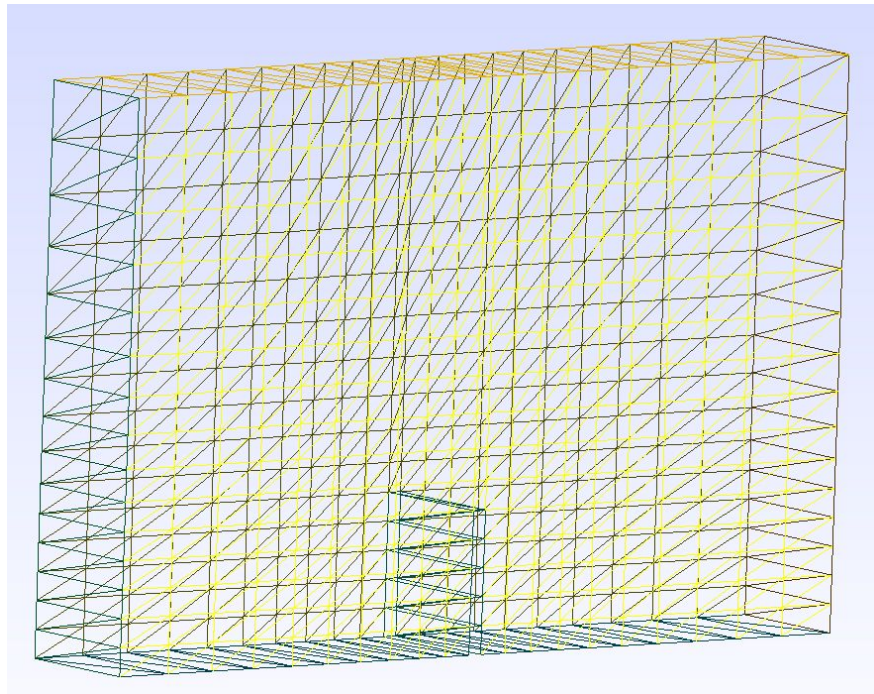


# More Differences



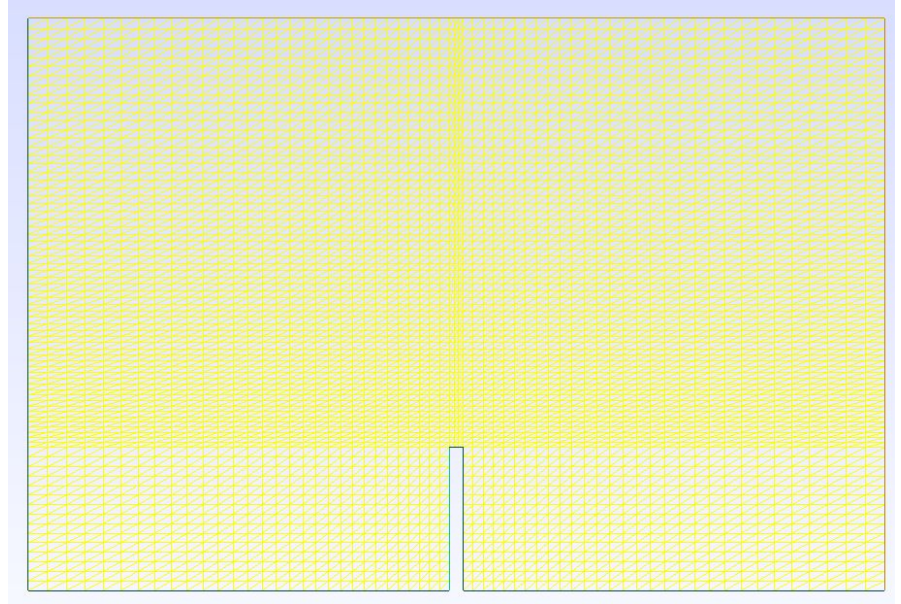
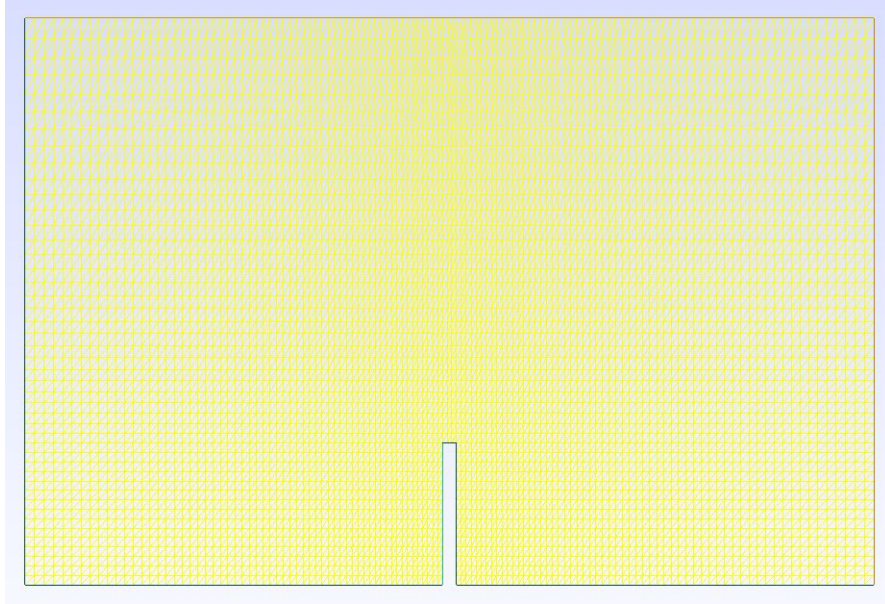


# Mesh Refinement

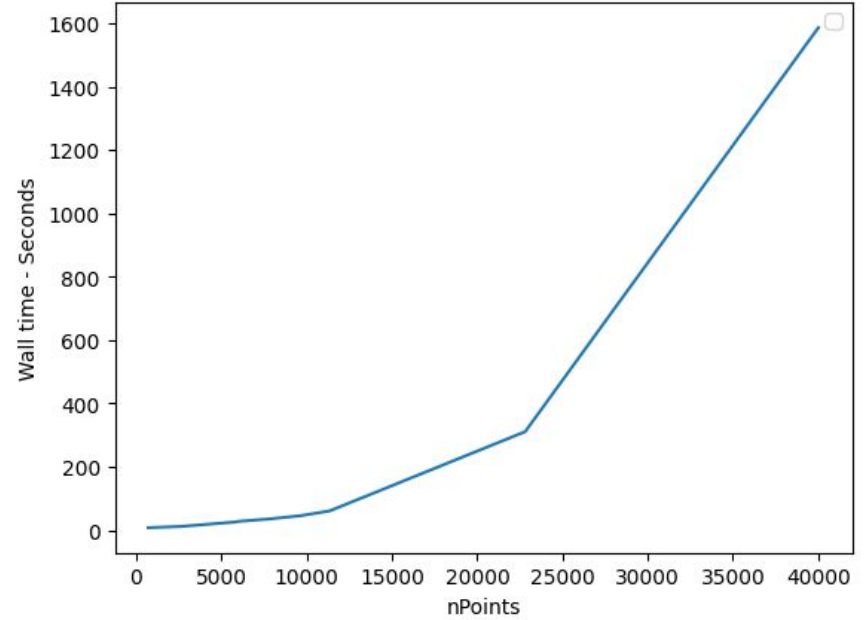
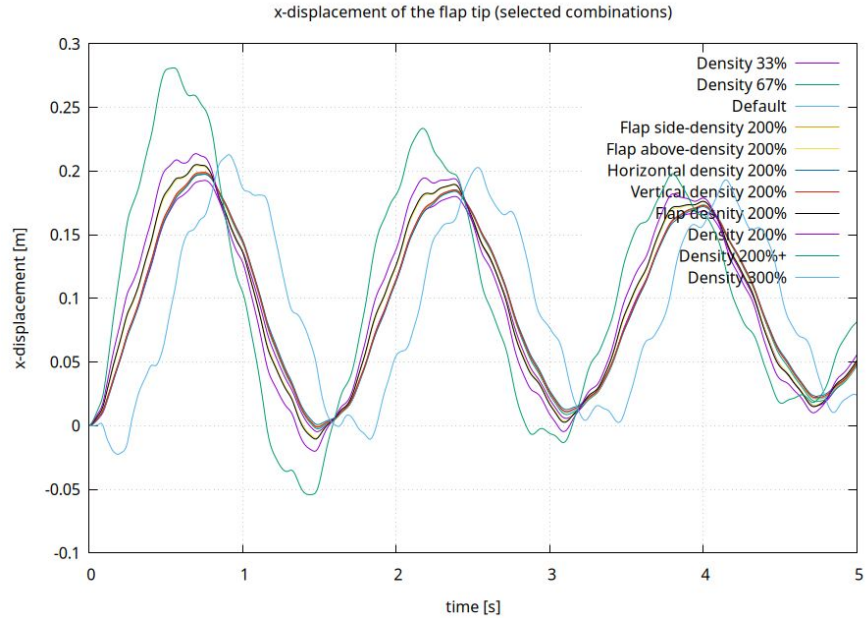




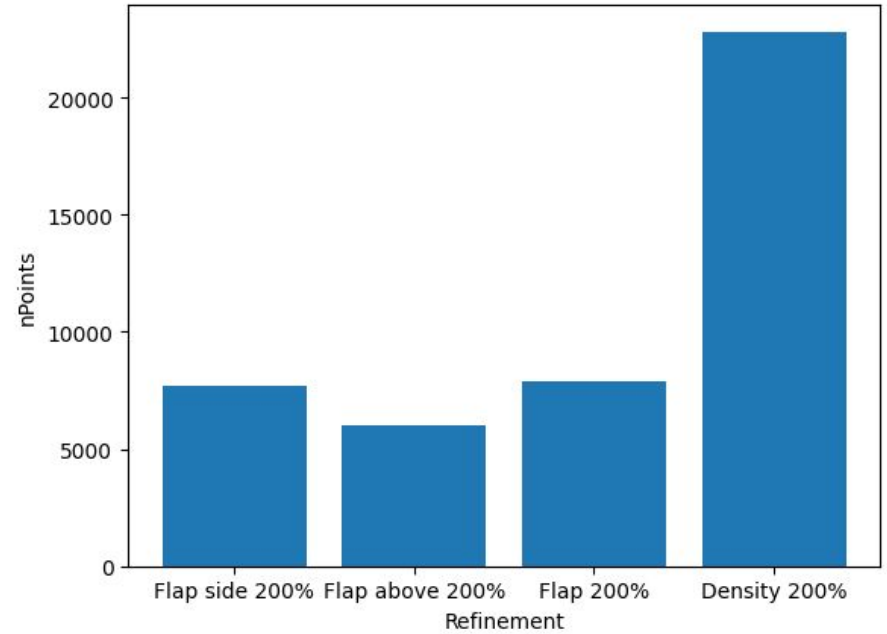
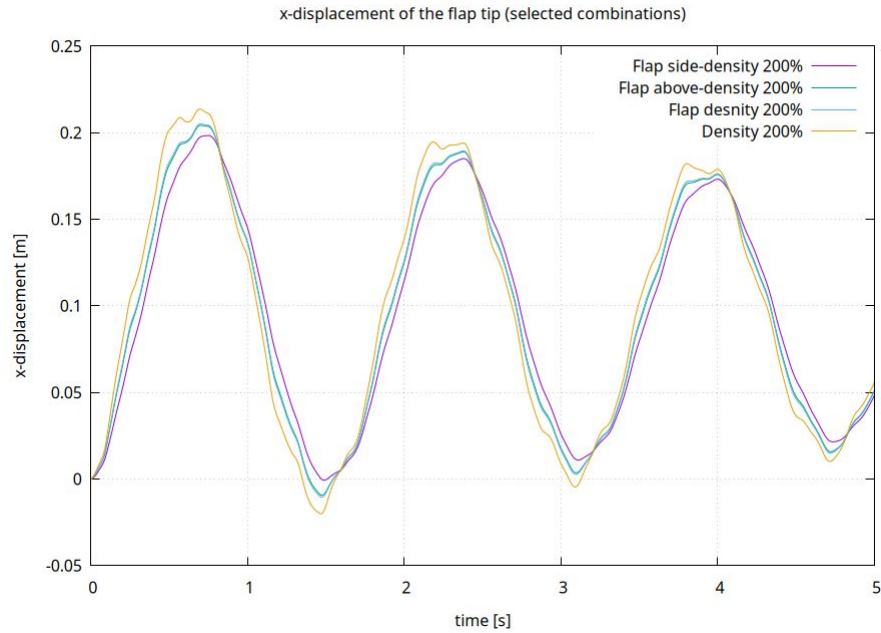
# Mesh Refinement



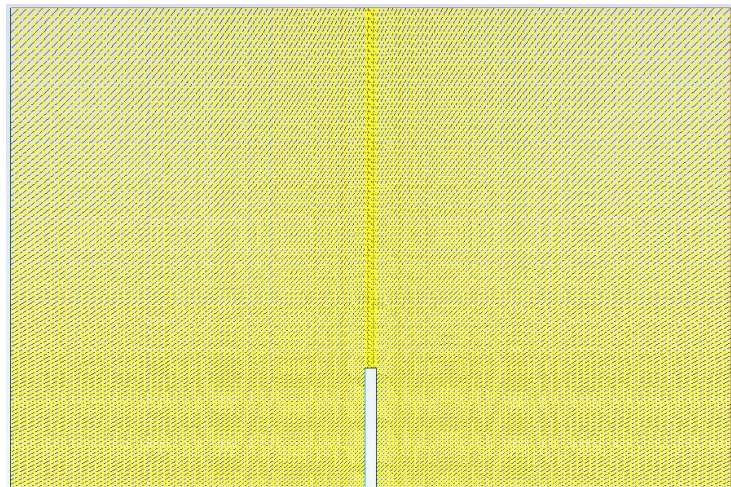
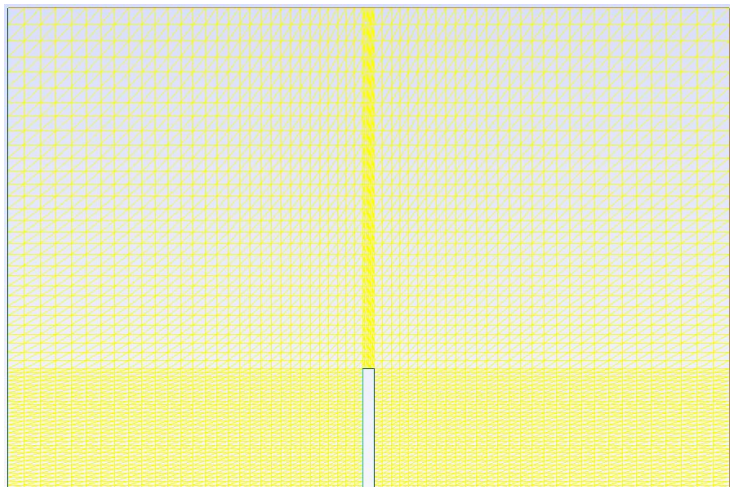
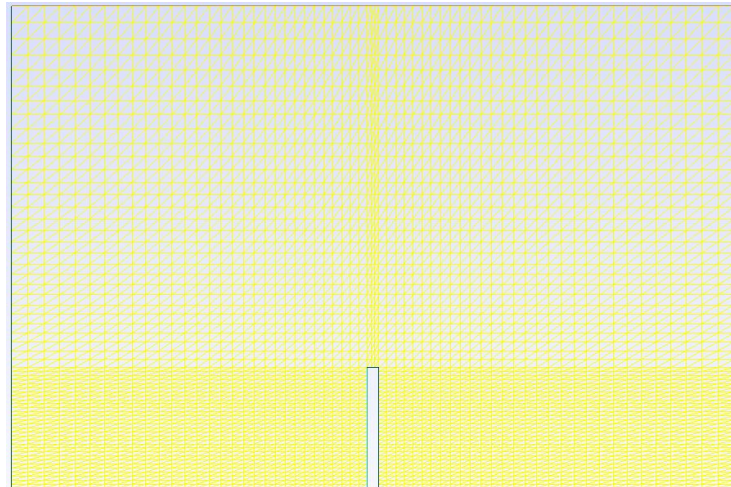
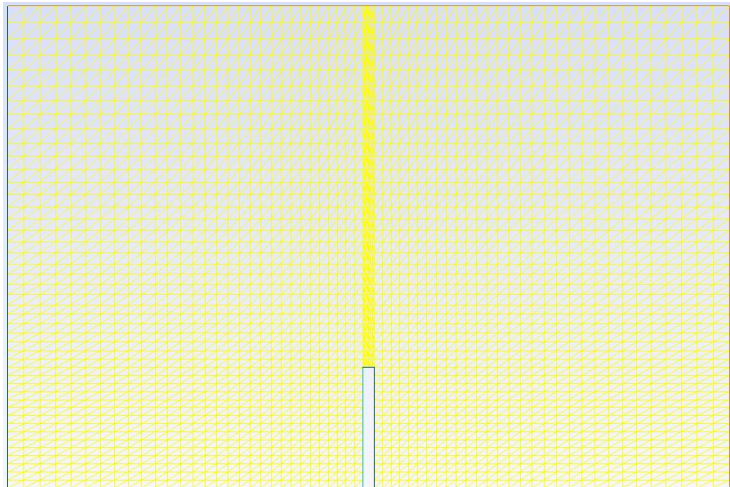
# Results



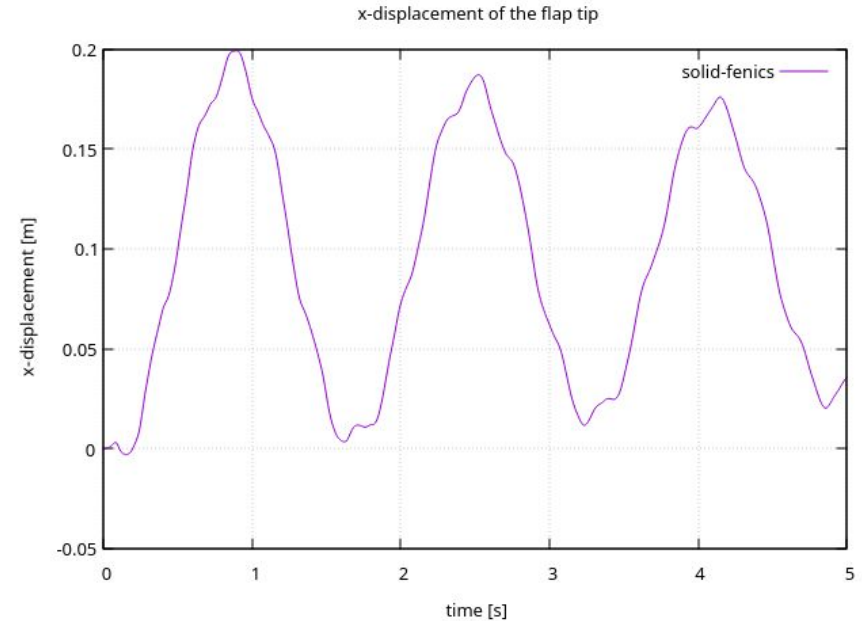
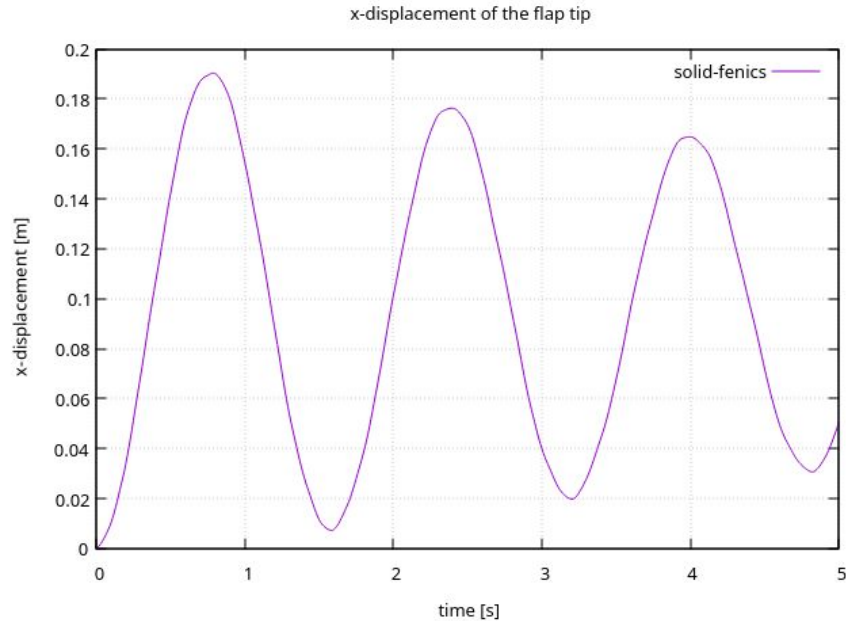
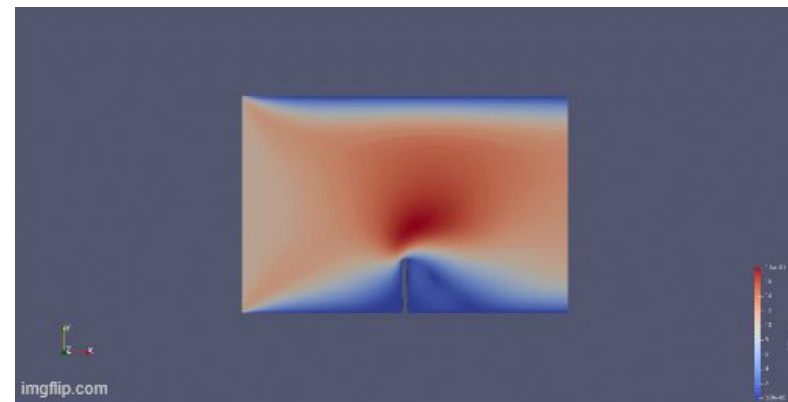
# Closer Look







# What about the solid mesh?





Thank you!